

Yong-chan Park

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About me

I am a postdoctoral researcher in Computer Science and Engineering at [Seoul National University](#) (SNU), advised by [Prof. U Kang](#) in [Data Mining Laboratory](#). My research interests include time-series analysis, tensor analysis, and anomaly detection.

Research vision and current work

My research lies at the intersection of machine learning and large-scale tensor analysis. I develop algorithms that make high-dimensional data representations both *faster* and *more interpretable*, with particular emphasis on streaming and irregular tensors common in web-scale applications. Recent projects include

- **Partial Fourier Transform:** a linear-time Fourier transform that accelerates spectral anomaly detection by $\times 20$ over prior art.
- **Data-Axis Transformer with Multi-Level Contexts:** the first end-to-end architecture that models inter-stock dependencies across markets for price-movement prediction.
- **PuzzleTensor:** a method-agnostic transformation that reduces the target rank of arbitrary tensors while preserving reconstruction fidelity.

Education

Postdoctoral Researcher Mar. 2026 – Present
- Computer Science and Engineering, SNU

Ph.D. Student Mar. 2020 – Feb. 2026
- Computer Science and Engineering, SNU

Bachelor of Science Mar. 2012 – Feb. 2019
- Department of Mathematical Sciences, SNU

Publications

- [1] **Yong-chan Park**, Jun-Gi Jang, and U Kang. “Fast and Accurate Partial Fourier Transform for Time Series Data.” KDD 2021
- [2] Jaemin Yoo, Yejun Soun, **Yong-chan Park**, and U Kang. “Accurate Multivariate Stock Movement Prediction via Data-Axis Transformer with Multi-Level Contexts.” KDD 2021
- [3] **Yong-chan Park***, Sangjun Son*, Minyong Cho, and U Kang. “DAO-CP: Data-Adaptive Online CP Decomposition for Tensor Stream.” PLOS One 2022 (*equal contribution)
- [4] Jun-Gi Jang, Jeongyoung Lee, **Yong-chan Park**, and U Kang. “Fast and Accurate Dual-Way Streaming PARAFAC2 for Irregular Tensors - Algorithm and Application.” KDD 2023
- [5] **Yong-chan Park**, Jongjin Kim, and U Kang. “Fast Multidimensional Partial Fourier Transform with Automatic Hyperparameter Selection.” KDD 2024
- [6] Jun-Gi Jang, **Yong-chan Park**, and U Kang. “Fast and Accurate PARAFAC2 Decomposition for Time Range Queries on Irregular Tensors.” CIKM 2024
- [7] JinGee Kim, **Yong-chan Park**, Jaemin Hong, and U Kang. “Accurate Stock Movement Prediction via Multi-Scale and Multi-Domain Modeling.” BigData 2024
- [8] SeungJoo Lee, **Yong-chan Park**, and U Kang. “Accurate Coupled Tensor Factorization with Knowledge Graph.” BigData 2024

- [9] **Yong-chan Park**, Kisoo Kim, and U Kang. "PuzzleTensor: A Method-Agnostic Data Transformation for Compact Tensor Factorization." KDD 2025
- [10] SeungJoo Lee, **Yong-chan Park**, and U Kang. "SwaGNER: Leveraging Span-aware Grid Transformers for Accurate Nested Named Entity Recognition." CIKM 2025
- [11] SeungJoo Lee, **Yong-chan Park**, and U Kang. "Offline and Online Coupled Tensor Factorization with Knowledge Graph." PLOS One 2025
- [12] Nam Kyu Kang, **Yong-chan Park**, and U Kang. "Fast and Accurate Temporal Super-Resolution via Residual-Aware Coupled Tensor Factorization." ICASSP 2026
- [13] **Yong-chan Park**, SeungJoo Lee, and U Kang. "Fast and Accurate Online Coupled Matrix-Tensor Factorization via Frequency Regularization." KDD 2026